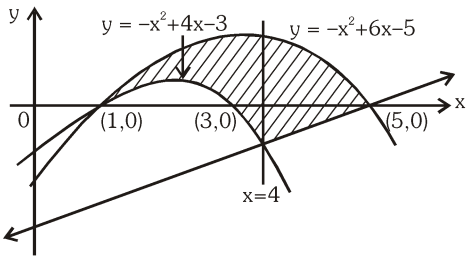
SOLUTION

SECTION–I

1. Ans. (C)



$$A = \int_1^4 \{(-x^2+6x-5)-(-x^2+4x-3)\}\,dx$$

$$+\int_4^5 (-x^2+6x-5)-(3x-15)dx = \frac{73}{6}$$

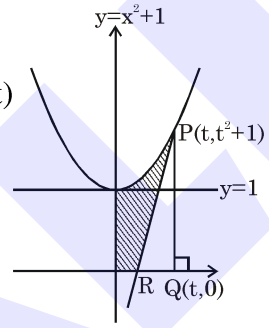
2. Ans. (B)

Equation of tangent

$y = (t^2+1) + 2t(x-t)$

$y = 0 \;;\; x = \dfrac{t^2-1}{2t}$

$R\left(\dfrac{t^2-1}{2t}, 0\right)$



$$= \int_0^t (x^2+1)\,dx - \Delta PQR = \frac{11}{3}$$

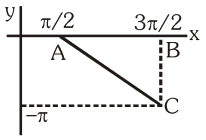
$= t^4 + 6t^2 - 44t - 3 = 0$

$(t-3)(t^3+3t^2+15t+1) = 0$

$t = 3$ is positive solution

$\Rightarrow$ x coordinate is 3

3. Ans. (D)



$$f(x) = \int_0^{\pi/2} [\cos t]\,dt + \int_{\pi/2}^{x} [\cos t]\,dt, \quad x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$$

$$\Rightarrow \quad f(x) = 0 + \int_{\pi/2}^{x} -1\,dt = \frac{\pi}{2} - x$$

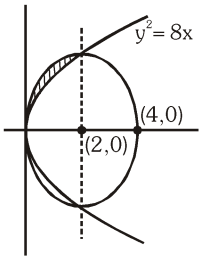
$$\Rightarrow \quad \text{Required area} = \frac{1}{2} \times AB \times BC = \frac{\pi^2}{2}$$

4. Ans. (D)

Area of shaded portion

$$= \frac{1}{4}(\pi.2.4) - \int_0^2 \sqrt{8x}\,dx$$

$$= 2\pi - \frac{16}{3}$$



5. Ans. (B)

$$\frac{1}{1+y^2}.\frac{dy}{dx} + 2x(\tan^{-1}y) = x^3$$

Put $\tan^{-1}y = z$

$$\therefore \quad \frac{1}{1+y^2}.\frac{dy}{dx} = \frac{dz}{dx}$$

$$\frac{dz}{dx} + (2x)z = x^3 \quad \Rightarrow \quad z.e^{x^2} = \frac{1}{2}\int 2e^{x^2}.x^3\,dx + c$$

Put $x^2 = t$

$$\Rightarrow \quad (\tan^{-1}y)\,e^{x^2} = \frac{1}{2}\int e^t.t\,dt + c$$

$$\Rightarrow \quad e^{x^2}(\tan^{-1}y) = \frac{1}{2}\{e^t.t - e^t\} + c$$

$$\Rightarrow \quad 2e^{x^2}\tan^{-1}y = x^2e^{x^2} - e^{x^2} + 2c$$

$$\Rightarrow \quad 2\tan^{-1}y = x^2 - 1 + 2ce^{-x^2}$$

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6. Ans. (C)

$(K+1)^2 e^{(K+1)x} - 4(K+1)e^{(K+1)x} + 4e^{(K+1)x} = 0$

$(K+1)^2 - 4(K+1) + 4 = 0$

$K + 1 = 2 \quad \Rightarrow \quad K = 1$

7. Ans. (C)

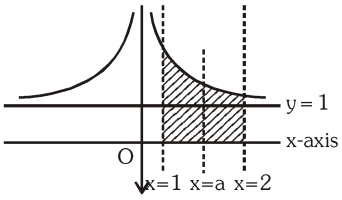
$$\left(\frac{dy}{dx}\right)^{\frac{1}{3}} = 4\frac{d^2y}{dx^2} + 7x$$

$$\left(\frac{dy}{dx}\right) = \left(4\frac{d^2y}{dx^2} + 7x\right)^3$$

$\Rightarrow$ order = 2

degree = 3

8. Ans. (D)



$$\Rightarrow x - \frac{1}{x}\Big|_1^a = x - \frac{1}{x}\Big|_0^2$$

$$\Rightarrow 2\left(a - \frac{1}{a}\right) = \frac{3}{2}$$

$$\Rightarrow 4a^2 - 3a - 4 = 0$$

$$\Rightarrow a = \frac{3 + \sqrt{73}}{8}$$

9. Ans. (A,B,D)

$$\frac{dy}{dx} = \frac{e^x}{\sqrt{1 - e^{2x}}}$$

$$\therefore \quad \int dy = \int \frac{d(e^x)}{\sqrt{1 - (e^x)^2}}$$

$$y = \sin^{-1}(e^x) + c \; = \cos^{-1}\sqrt{1 - e^{2x}} + c$$

$$y = \frac{\pi}{2} + c - \cos^{-1}(e^x) \Rightarrow y = k - \cos^{-1}e^x$$

10. Ans. (A,B)

$$\int \frac{bdy}{y^2 + b^2} + \int \frac{adx}{x^2 + a^2} = 0$$

$$\Rightarrow \quad \tan^{-1}\left(\frac{x}{a}\right) + \tan^{-1}\left(\frac{y}{b}\right) = c$$

Apply tan on both sides

$$\frac{\frac{x}{a} + \frac{y}{b}}{1 - \frac{xy}{ab}} = k \quad \Rightarrow bx + ay = k(ab - xy)$$

11. Ans. (B,C,D)

$d(xy) = dx$

integrate both sides $xy = x + c$

$$c = 1 \quad \Rightarrow \quad y = 1 + \frac{1}{x}$$

(1) Non periodic function

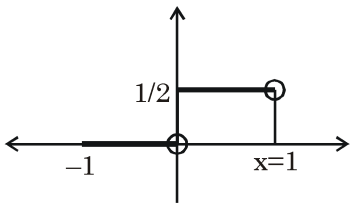
(2) $f(x_1) = f(x_2) \Rightarrow \quad x_1 = x_2$

(3) Domain $R - \{0\}$

(4) Neither even nor odd

12. Ans. (A,B,C)

$2\{y\} = [x] + 1 \qquad \ldots\ldots(1)$



$\because \quad 0 \le y < 1 \quad \Rightarrow \{y\} = y$

$\therefore \quad (1) \Rightarrow \quad 2y = [x] + 1$

$0 \le [x] + 1 < 2$

$-1 \le [x] < 1$

but $[x]$ is an integer

$\Rightarrow \quad [x] = -1 \quad \Rightarrow \quad -1 \le x < 0$

$[x] = 0 \quad \Rightarrow \quad 0 \le x < 1$

when $x \in [-1, 0) \quad \Rightarrow \quad y = 0$

when $x \in [0, 1) \quad \Rightarrow \quad y = \frac{1}{2}$

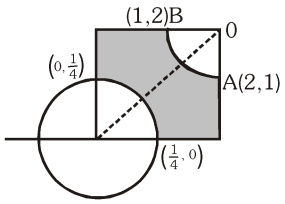
$$\therefore \text{ Area} = \frac{1}{2}$$

13. Ans. B,C,D

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SECTION–II

1. Ans. 4.00



$xy = 2$ is rectangular hyperbola & $8(x^2+y^2) \geq 2$ is exterior of circle; shaded portion is given by figure.

Area of portion OAB $= 2 - \int_1^2 \frac{2}{x}dx = (2 - \ell n4)$

Total area $= 4 - \left(\frac{\pi}{16} + 2 - \ell n4\right)$

$2 + \ell n4 - \frac{\pi}{16}$

2. Ans. 1.00

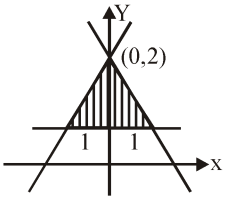
$x^2y - x^2 - y^3 + y^2 + 4y^2 - 4y - 4y + 4 = 0$

$\Rightarrow x^2(y-1) - y^2(y-1) + 4y(y-1) - 4(y-1) = 0$

$\Rightarrow (y-1)(x^2 - y^2 + 4y - 4) = 0$

$\Rightarrow$ $y = 1$ or $x^2 = (y-2)^2$

$\Rightarrow$ $y = 1$, $x = y - 2$ or $x = 2 - y$



Shaded area $= \frac{1}{2} \times 2 \times 1 = 1$ sq. units

3. Ans. 2.00

$(\tan y + y\sec^2 y)dy = e^x(\sin x + \cos x)dx$

$y\tan y = e^x \sin x + C$

(0,0) gives $C = 0$

$y\tan y = e^x \sin x$

$\pi \tan \pi = e^{x_1} \sin x_1$

$\Rightarrow \sin x_1 = 0 \Rightarrow x_1 = 2\pi$

$\frac{x_1}{\pi} = 2$

4. Ans. 7.00

$y = ax^p + bx^{-q}$

$\Rightarrow y' = apx^{p-1} - bqx^{-q-1}$

$\Rightarrow x^2y' = apx^{p+1} - bqx^{1-q}$

$\Rightarrow x^2y'' + 2xy' = (p+1)\, apx^p - bq(1-q)x^{-q} = 12y$

$\Rightarrow p(p+1) = 12$ & $q(q-1) = 12$

$\Rightarrow p = 3$ & $q = 4$

5. Ans. 12.00

$d(x^2y) = d(xy) + d(xy^2)$

$\Rightarrow x^2y = xy + xy^2 + c$

at $x = 1$, $y = 1$ $\Rightarrow$ $c = -1$

$\Rightarrow x^2y = xy + xy^2 - 1 \Rightarrow y = -y - y^2 - 1$

$\Rightarrow y^2 + 2y + 1 = 0$ $\Rightarrow$ $y = -1$

$\therefore$ $12|y(-1)| = 2$

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